"Pre-Registered Protocol — Front-Consistency Veto (Radio Air-Shower Arrays)"

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Pre-Registered Protocol

Front-Consistency Veto for RFI Rejection in Radio Air-Shower Arrays: Full Simulation Study

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Abstract

Radio air-shower arrays trigger on nanosecond broadband pulses in a crowded RF environment,

and the residual that survives notch filtering and offline direction-of-arrival cuts is increasingly the impulsive, near-field kind. This protocol pre-registers the full simulation study of a trigger-level rejection layer based on a simple physical asymmetry: a shower arrives as a nearly planar, slightly curved front, while a point-source RFI transmitter at finite distance carries a spherical front whose curvature radius equals its distance. The gate fits a plane and curvature to each candidate’s inter-antenna delays and accepts only events whose fitted curvature radius exceeds a pre-specified floor. An exploratory skeleton study (seed 684, parameterized models) established the mechanism and exposed two findings this protocol encodes: the gate boundary is soft, with reliable rejection inside roughly 3 km grading into a quantified blind region beyond about 12 km, and rejection is statistically independent of RFI waveform

class. The full study replaces the skeleton's idealized models with CoREAS shower pulses, published RFI environments, and AERA-class array geometry, under five pre-registered falsifiers with Wilson confidence intervals required on every reported rate. This document fixes all thresholds before the full data exist.

Keywords: radio detection of cosmic rays; RFI; trigger; curvature; CoREAS; pre-registration; AERA; BEACON; GRAND

1. Introduction and lineage

   The concept note for this method is archived and citable [1]; the reader is referred there for the motivation and the division of labor with existing methods. What matters here is what has happened since that note was locked. A skeleton simulation (v0.2, seed 684, seven-antenna hexagonal array, parameterized fronts) was built and run under the same Run-

All discipline as the companion swarm validation [2,3]. It was built to answer one question — does the front-consistency mechanism separate showers from point-source RFI at all? — and it answered two more in passing.

First, the mechanism is curvature, and only curvature. A distant transmitter is indistinguishable from a shower in delay geometry, because both are planar across a small array; what betrays a near-field source is the spherical sagitta of its wavefront. Second, the gate boundary is soft. The pre-specified curvature floor sits at 5 km, but fitted curvature spreads around the true source distance, so rejection does not step at the floor — it holds at 100% well inside 1.5 km, fades through the 3–6 km decade, and reaches the accidental floor beyond about 12 km. A third observation, quieter but load-

bearing: the three canonical RFI waveform classes gave statistically identical rejection at every distance. The spatial layer does not care what the interference looks like in time; that is the temporal layer's job.

This protocol exists because those findings change what must be locked. A gate with a soft boundary needs its floor, its calibration procedure, and its distance-binning fixed in advance, or the full study's numbers will be tuned after the fact. Everything tunable is fixed below. The only open fields are operational parameters collected in the lock-time table (§9).

2. Objective and scope

   The objective is to validate the front-consistency veto under realistic simulation and to map its operating envelope with confidence intervals. Four claims are tested:

(i) near-field impulsive RFI is rejected at trigger level with a bounded, distance-quantified blind region; (ii) shower retention meets a fixed floor with no directional or energy bias; (iii) all thresholds derive from a calibration subset and transfer unchanged; (iv) the gate tolerates realistic array degradation.

Scope: this protocol covers simulation only. Application to any collaboration's data is a separate study with its own registration, proposed only after F1–F5 pass here.

3. Definitions

Array: seven antennas, hexagon of radius 150 m plus center (skeleton geometry retained for comparability). The CoREAS upgrade arm uses published AERA-class station positions; F5 degradation tests apply to both.

Candidate event: at least three antennas

above the energy threshold within the trigger window (the comparator is the pure energy trigger, unchanged).
Gate, stage one: least-squares plane fit to the candidate's inter-antenna delays; post-fit residual RMS in ns.
Gate, stage two: curvature extraction against squared transverse distance to the fitted axis; fitted curvature radius RC = 1/(2κc). Accept only if residual RMS < THR and RC > RC_MIN, else reject (flag as RFI-candidate).
RC_MIN: 5 km. Locked by this protocol; changeable only by a new versioned protocol.
THR: the 95th percentile of stage-one residual over the calibration shower subset. The procedure is locked here; the value is computed once, from the calibration arm only, and recorded.
Distance bins (locked): 0.5–1.5, 1.5–3, 3–6, 6–12, 12–25, 25–50 km.

Confidence reporting: every rate in the results paper carries a Wilson 95% score interval. Point estimates without intervals are not reportable.

4. Protocol core

   Arms. (a) Showers: CoREAS-simulated pulses for zenith 0–60°, uniform azimuth, primary energy log-uniform 1017–1018.5 eV, with published lateral distribution and a single-channel antenna response model recorded in the lock-time table. (b) RFI: three classes (carrier-like, periodic, random-impulsive) drawn from a published RFI environment census where available, distance distribution log-uniform 0.5–50 km. (c) Controls: pure energy trigger on both populations (the comparator); calibration/test splits drawn once from the seed.

   Calibration and transfer (F4). THR is

computed on a calibration shower subset (30% of shower events) and applied unchanged to the test set. RC_MIN never touches calibration data. Any post-hoc adjustment voids the corresponding falsifier and is disclosed under §8.

Statistics. Rejection and retention rates with Wilson intervals; KS tests of retained-sample zenith, azimuth, and log-energy distributions against the triggered input; per-bin counts reported alongside every rate.

5. Pre-registered falsifiers

Failure of any falsifier at any arm voids the corresponding claim; deviations are reported, not repaired.

ID

Claim protected

Criterion (fixed by this protocol)

F1

Near-field RFI rejection

Rejection ≥ 99% with Wilson lower bound ≥ 95% in the 0.5–1.5 km bin, per class; full distance curve reported; blind region defined as the nearest bin whose lower bound falls below 10%

F2

Shower retention

Retention ≥ 90% with lower bound ≥ 88% against the pure energy comparator

F3

No selection bias

KS p ≥ 0.05 on zenith, azimuth, log-energy of retained vs triggered; no 10°-azimuth bin deviates in efficiency by more than 5 points

F4

Threshold transfer

THR from calibration subset only, applied unchanged; RC_MIN untouched by calibration; binary

F5
Degraded-array robustness
With one and with two antennas randomly excluded, F2 retention within 3 points of full-array retention and F1 conclusions unchanged in the 0.5–1.5 km bin

6. Exploratory anchors (skeleton v0.2, seed 684)
   The following numbers derive from the parameterized skeleton and are recorded here to show that the locked criteria are neither optimistic nor punitive: rejection at <1.5 km was 100% (lower bound 96.7%); the 3–6 km bin measured 68.6% [64.9–72.1]; the 6–12 km bin 21.6% [18.3–25.4]; the blind region set in

beyond 12 km at the ~5% accidental floor. Shower retention measured 94.7% [93.7–95.5] with no measurable bias; one- and two-antenna degradation cost under three points. The full study's falsifier thresholds are set against these values with margin, and the exploratory status of the skeleton is stated wherever its numbers appear: they motivated the locks; they are not evidence for them.

2. Known failure modes, declared in advance
First, far-field RFI is planar and belongs to the direction-catalog layer, not this gate; the blind region is a design boundary, quantified by F1, not an embarrassment to be minimized. Second, fitted-curvature spread makes the boundary soft; the distance curve, not the RC_MIN line, is the deliverable. Third, if the CoREAS arm's antenna response model proves materially different from the skeleton's,

THR recomputed under the locked procedure may shift; the procedure, not any number, is what is frozen. Fourth, a transmitter physically collocated with the reconstructed shower axis mimics a front; no delay-only gate can see this, and the protocol does not claim to.

3. Versioning and deviation policy
   This document is immutable once timestamped. Amendments are new versions with dated changelogs, issued before the affected data are generated. Deviations during execution are logged and disclosed with statistical consequences; selective deviation voids pre-registration status for the affected claim.

4. Lock-time parameter table
   Fields not fixed by any prior document must be completed and initialed before the first full-study run.

Parameter

Value

1:CoREAS shower file set / provenance

2:Antenna response model and citation

3:RFI environment census citation

4:AERA-class station position source

5:Energy threshold (comparator) value

6:Timing jitter model (σ, distribution)

7:THR value, as computed from calibration arm

8:Calibration/test split realization (seed note)

Declarations

Funding: None (independent research).
Competing interests: None. Data and code availability: No data accompany this protocol. The full study will run in a Run-All notebook seeded from the skeleton’s lineage; notebook, results, and figures will be archived with the results publication. Author contributions: N.K. designed the gate, ran the skeleton study, and wrote this protocol. Correspondence: Nasirk684@gmail.com

Initial pre-registration, incorporating skeleton v0.2 exploratory findings.

References

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CoREAS simulation reference (to be completed with full citation at study submission).

AERA self-trigger performance reference (indicative).

LOFAR radio-detection trigger literature (indicative).

BEACON / GRAND trigger-design literature (indicative).